BST Solutions

Solution 1:

Time Complexity : o(n)
Space Complexity: o(n)

```
import java.util.*;

class Solution{

    static class Node{
    int val;
    Node left, right;
    };

    static Node newNode(int item){
    Node temp = new Node();
    temp.val = item;
    temp.left = temp.right = null;
    return temp;
    }


    static int sum = 0;

    static int rangeSumBST(Node root, int low,
                                                int high){
        if (root == null)
        return 0;

        Queue<Node> q = new LinkedList<Node>();
        q.add(root);

        while (q.isEmpty() == false){

            Node curr = q.peek();
            q.remove();
            if (curr.val >= low &&
                    curr.val <= high){
                    sum += curr.val;
            }
```

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```
                if (curr.left != null &&
                        curr.val > low) q.add(curr.left);
                if (curr.right != null &&
                        curr.val < high)
                        q.add(curr.right);
            }
            return sum;
    }

    static Node insert(Node node, int data){

            if (node == null)
                    return newNode(data);
            if (data <= node.val)
                    node.left = insert(node.left,
                                                    data);
            else
                    node.right = insert(node.right,
                                            data);
            return node;
    }

    public static void main(String[] args){

            Node root = null;
            root = insert(root, 10);
            insert(root, 5);
            insert(root, 15);
            insert(root, 3);
            insert(root, 7);
            insert(root, 18);

            int L = 7, R = 15;
            System.out.print(rangeSumBST(root, L, R));
    }
    }
```

Solution 2:

Time Complexity : o(h)
Space Complexity: o(1)

```
class Solution{
```

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```
        static int min_diff, min_diff_key; static class Node{
        int key;
        Node left, right;
};

static Node newnode(int key){
        Node node = new Node();
        node.key = key;
        node.left = node.right = null;
        return (node);
}

         static void
  maxDiffUtil(Node ptr, int
      k){ if (ptr == null)
                return ;

        if (ptr.key == k){
                min_diff_key = k;
                return;
        }

        if (min_diff > Math.abs(ptr.key - k)){
                min_diff = Math.abs(ptr.key - k);
                min_diff_key = ptr.key;
        }

        if (k < ptr.key)
                maxDiffUtil(ptr.left, k);
        else
                maxDiffUtil(ptr.right, k);
}

static int maxDiff(Node root, int k){
        min_diff = 999999999;
        min_diff_key = -1;
        maxDiffUtil(root, k);
        return min_diff_key;
}
```

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```
public static void main(String args[]){
        Node root = newnode(9); root.left = newnode(4);
        root.right = newnode(17);
        root.left.left = newnode(3);
        root.left.right = newnode(6);
        root.left.right.left = newnode(5);
        root.left.right.right = newnode(7);
        root.right.right = newnode(22);
        root.right.right.left = newnode(20);
        int k = 18;
        System.out.println( maxDiff(root, k));

}
}
```

Solution 3 :

Time Complexity : o(n)
Space Complexity: o(h)

```
import java.io.*;

class Node {
        int data;
        Node left, right;
        Node(int x)
        {
                data = x;
                left = right = null;
        }
}

class Solution {

        static int count = 0;
        public static Node insert(Node
                root, int x) { if (root ==
                null)
                        return new Node(x);
```

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```
            if (x < root.data)
                root.left = insert(root.left, x);
            else if (x > root.data) root.right = insert(root.right, x);
            return root;
        }

        public static Node kthSmallest(Node root, int k){

            if (root == null)
                return null;

            Node left = kthSmallest(root.left, k);
            if (left != null)
                return left;
            count++;
            if (count == k)
                return root;
            return kthSmallest(root.right, k);
        }

        public static void printKthSmallest(Node root, int k){
            Node res = kthSmallest(root, k);
            if (res == null)
              System.out.println("There are less than k nodes in the BST");
            else
                System.out.println("K-th Smallest Element is " + res.data);
        }

        public static void main(String[] args){
            Node root = null;
            int keys[] = { 20, 8, 22, 4, 12, 10, 14 };
            for (int x : keys)
                root = insert(root, x);
            int k = 3;
            printKthSmallest(root, k);
        }
}
```

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Solution 4 :

Time Complexity : o(n1+n2)
Space Complexity: o(h1+h2)

```
import java.util.Stack;
public class Solution {

	static class Node {
		int data;
		Node left, right;

		public Node(int data) {
			this.data = data;
			left = null;
			right = null;
		}
	}

	static Node root1;
	static Node root2;
	static int countPairs(Node root1, Node root2,
	                                                    int x)
	{
		if (root1 == null || root2 == null)
			return 0;
		Stack<Node> st1 = new Stack<>();
		Stack<Node> st2 = new Stack<>();
		Node top1, top2;

		int count = 0;
		while (true) {
			while (root1 != null) {
				st1.push(root1);
				root1 = root1.left;
			}
			while (root2 != null) {
				st2.push(root2);
				root2 = root2.right;
```

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```
                }
                if (st1.empty() || st2.empty()) break;

                top1 = st1.peek();
                top2 = st2.peek();
                if ((top1.data + top2.data) == x) {
                        count++;
                        st1.pop();
                        st2.pop();
                        root1 = top1.right;
                        root2 = top2.left;
                }

                else if ((top1.data + top2.data) < x) {
                        st1.pop();
                        root1 = top1.right;
                }

                else {
                        st2.pop();
                        root2 = top2.left;
                }
        }
        return count;
}

public static void main(String args[])
{
        root1 = new Node(5);
        root1.left = new Node(3);
        root1.right = new Node(7);
        root1.left.left = new Node(2);
        root1.left.right = new Node(4);
        root1.right.left = new Node(6);
        root1.right.right = new Node(8);

        root2 = new Node(10);
        root2.left = new Node(6);
        root2.right = new Node(15);
        root2.left.left = new Node(3);
```

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```
        root2.left.right = new Node(8); root2.right.left = new Node(11);
        root2.right.right = new Node(18);

        int x = 16;
        System.out.println("Pairs = "
                + countPairs(root1, root2, x));
    }
}
```

Solution 5 :

Time Complexity : o(n)
Space Complexity: o(n)

```
class Solution {

static class Node{
    Node left;
    Node right;
    int data;

    Node(int data){
        this.data = data;
        this.left = null;
        this.right = null;
    }
};

static class Info{
    int max;
    int min;
    boolean isBST;
    int sum;
    int currmax;

    Info(int m,int mi, boolean is,
        int su, int cur) {
```

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```
        max = m;
            min = mi;
            isBST = is;
            sum = su;
            currmax = cur;
        }
        Info(){}
};

static class INT{
        int a;
}

static Info MaxSumBSTUtil( Node root, INT maxsum){
        if (root == null)
            return new Info( Integer.MIN_VALUE,
                            Integer.MAX_VALUE, true, 0, 0 );
        if (root.left == null && root.right == null){
            maxsum.a = Math.max(maxsum.a, root.data);
            return new Info( root.data, root.data,
                            true, root.data, maxsum.a );
        }

        Info L = MaxSumBSTUtil(root.left, maxsum);
        Info R = MaxSumBSTUtil(root.right, maxsum);

        Info BST=new Info();
        if (L.isBST && R.isBST && L.max < root.data &&
                                        R.min > root.data) {

            BST.max = Math.max(root.data, Math.max(L.max, R.max));
            BST.min = Math.min(root.data, Math.min(L.min, R.min));

            maxsum.a = Math.max(maxsum.a, R.sum +
            root.data + L.sum); BST.sum = R.sum + root.data +
            L.sum;
            BST.currmax = maxsum.a;

            BST.isBST = true;
            return BST;
```

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```
        }
        BST.isBST = false;
        BST.currmax = maxsum.a;
        BST.sum = R.sum + root.data + L.sum;
                                                    return BST;
}


 static int MaxSumBST( Node root){
        INT maxsum = new INT();
        maxsum.a = Integer.MIN_VALUE;
        return MaxSumBSTUtil(root,
maxsum).currmax; }

 public static void main(String args[]){
        Node root = new Node(5);
        root.left = new Node(14);
        root.right = new Node(3);
        root.left.left = new Node(6);
        root.right.right = new Node(7);
        root.left.left.left = new Node(9);
        root.left.left.right = new Node(1);

        System.out.println( MaxSumBST(root));
}
}
```

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